

Discourse Analysis: Analyzes the

across multiple sentences in a document.

⇒ unlike statistical methods, NLP identifies the actual meaning (concepts) present in the document.

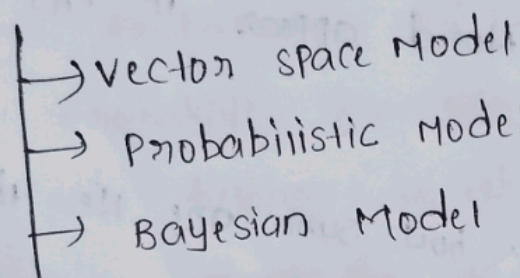
Automatic indexing



indexing by term

↓
statistical techniques

↓
Natural language techniques



indexing by concept:

instead of indexing exact words the slm indexes concepts.

Different words with the same meaning are grouped together.

Ex:-

car
Automobile
vehicle

All represent the same concept.

Multimedia indexing: multimedia indexing is the process of automatically or manually indexing multimedia data, such as images, audio, video, and graphics using their features, meta data on content to enable efficient searching & retrieval.

Data Structures:

Introduction to Data Structures:

A data structure is a method of organizing and storing data so that it can be processed, searched, updated and retrieved efficiently.

In an information retrieval system data structures organize both the documents and their searchable terms to enable fast and accurate retrieval.

Need for data structures:

Data structures are required because an IRS stores thousands on millions of documents, without proper organization.

- * Searching becomes slow
- * Storage requirements increase
- * Retrieval accuracy decreases
- * Updating documents becomes difficult.

Therefore efficient data structures are essential for fast searching and retrieval.

The 2 major data structures are:

- 1) Document structure
- 2) Search structure.

Document structure: The document structure stores and manages the retrieved received documents in their normalized form.

This function is performed by the "Document manager"

Functions of Document manager:

- * Stores original documents
- * Assigns document identifiers

- * main-lain document database
- * Retrieves complete documents
- * updates & deletes documents
- * Provides documents after search.

Stored information.

- Document ID
- Document Name
- Document Content
- file location
- Metadata

Search structure:

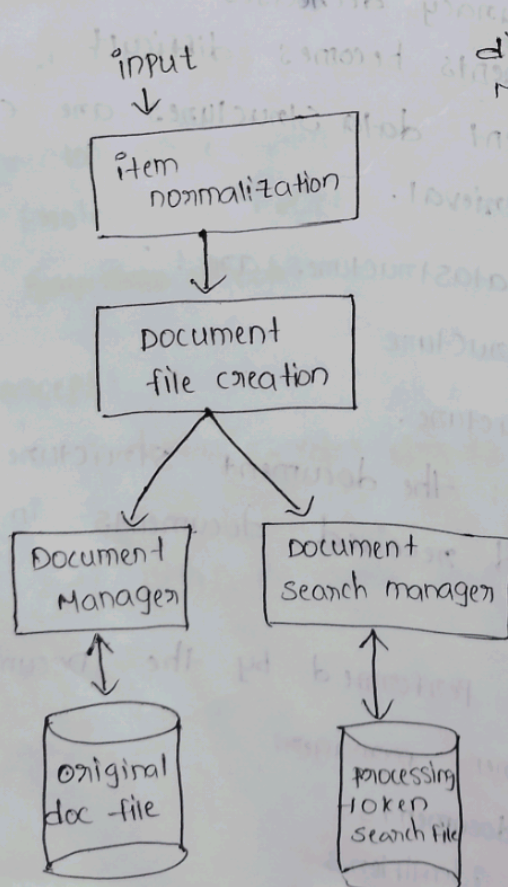
the second DS stores searchers information extracted from documents

it contains tokens, keywords, indexes, processing terms.

this structure is managed by the "document search manager"

functions.

- token generation
- index creation
- searching
- Ranking
- Matching query terms.



Document input:

Documents are received from users or databases.

ex:- IRS is an important subject.

Item normalization: Before storing documents are normalized, normalization converts different representations into one standard form.

Operations include:

⇒ Convert UPPercase to lowercase

⇒ remove punctuation

⇒ Remove special symbols

⇒ Remove extra spaces

⇒ Remove stopwords

⇒ -APPLY stemming.

ex:- Original

computers,

COMPUTERS,

normalized

computer.

Document file search: creation:

The normalized document is stored permanently

ex:- Document 1

Document 2

Document 3

Document Manager:

stores complete original documents.

Responsibilities:

* maintain document database

* Retrieve original documents

* Update documents

* delete documents

* Manage storage.

Document search Manager:

creates searchable indexes

stores → tokens, stems, inverted indexes.

Search structures.

Search file: contains searchable processing tokens instead of full documents when the user searches only this file is searched first.

Document retrieval: After matching terms, document IDs are sent to the Document Manager.

The Document Manager retrieves complex documents.

Search process

User query



Tokenization



Search processing tokens



Matching document IDs



Retrieve original documents.

Advantages:

- * fast searching
- * reduced storage
- * efficient indexing
- * easy document retrieval
- * Better search performance.

⇒ Stemming:

Stemming is the process of reducing word to different forms of a word to a common root word by removing prefixes or suffixes it is widely used in IRS to improve search efficiency & recall

Example:-

original words	stem
playing, played, plays	play
computing, computer, computation	comput

calculate, calculated,

calculating

calculate

connect, connected,

connect.

connecting, connection

the concept of stemming has been used in information retrieval (IR) systems since the 1960s.

the original purpose of stemming was to reduce the number of unique words stored in the system.

* to save storage space.

* to improve search efficiency

* to increase recall during information retrieval.

Today, storage and processing power are less of a problem, but stemming is still used because it improves retrieval effectiveness.

ex:- Suppose a user search for

"calculate"

Relevant documents may contain

* calculate

* calculation

* calculates

* calculations

* calculating

Without stemming the system retrieves only docs containing "calculate"

With stemming all words are converted to calculate

Hence all relevant documents are retrieved.

Need for stemming:

1) Reduces dictionary size

2) improves recall

3) Reduces search time

4) Handles morphological variations.

Reduces dictionary size: instead of storing connect, connects,

-d, connecting, connection the system stores only 'connect'.

improves recall: More relevant documents can be retrieved.
Reduces search time: Searching one stem is faster than searching multiple word variations.

Handles Morphological variations:

Different forms of a word are treated as a single concept.

* Stemming process: A stemming algorithm maps multiple morphological forms of a word into a single stem.

Example:-

Words	stem
compute	comput
computer	comput
computing	comput
computation	comput

the term represents the basic concept shared by all variants.

* Recall & Precision in stemming

Recall: Recall measures how many relevant documents are retrieved.

Precision: Precision measures how many retrieved documents are actually relevant.

⇒ Effect of stemming :-

Recall generally increases

Precision may decrease

ex:- Query "bank"

retrieved documents: River bank,
State bank,
Banking system.

* Not all documents may be relevant.

→ Hence, Recall increases but Precision may decrease.

Major stemming approaches:

* Porter stemming

* Dictionary lookup stemmers

* Successor variety stemmers.

Porter Stemming algorithm:

The Porter stemmer/stemming algorithm is one of the most popular based stemming algorithms used in IRS. It was introduced by Martin Porter in 1980.

It removes prefixes & suffixes from English words to obtain their stem (root form).

It is widely used in information retrieval systems (IRS) to improve indexing & searching.

Main concept: Instead of storing every form of a word separately, the algorithm stores only the stem.

Why Porter algorithm: Without stemming searching for connect will retrieve only "connect".

After stemming: Searching for connect will retrieve.

- connect
- connected
- connecting
- connectivity
- connection.

Thus, → increases recall

→ reduces index size

→ improves search speed

original word



check rule



Remove suffix



Apply next rule



final stem.

* Measure (M):

The most important concept in Porter algorithm is Measure (M)

It represents the no. of vowel-consonant (vc) sequence in a word

→ A stem consists of (c)vcvcv(c)

c - consonant

v - vowel

() → optional

ex:- 1) -lnee
 -ln = c
 ee = v
 cv
 M=0

2) -lnooble
 -ln = c
 ou = v
 bl = c
 e = v
 cvcv
 M=1

3) cvcv cv
 vc = 2
 M = 2

⇒ stem conditions:- the porter algorithm uses stem conditions to check whether a suffix. there are 5 conditions remove rule is valid. these conditions prevent incorrect stemming.

Preserve meaning.

* X Means stem ends with X

-full root words & improve the accuracy of doc retrieval in IRS.

ex:- Caness
 ends with
 Condition satisfied.

2) * Vx

stem contains atleast one vowel

ex:- playing
 play
 contains vowel
 Condition - true.

3) * d : stem ends with double consonants.

ex:- fail
 press
 hopp
 stopp

ll, ss, pp

are double consonants

(after remaining suffixes like ing or ed the algorithm often removes stem ends with one of the double consonant)

4) * O : Means stem ends with consonant

vowel

Consonant.

ex:- hop
 h = c

O = v

P = c

cvc condition satisfied.

conflation → Process of grouping different forms of a word into a single common root (stem)

Rules: Step 1a Rules

sses
↓
ss

ex:- stresses → stress

ies → i ex Ponies → Pon

ss → ss - no change ex:- glass → glass

Rule: s → NULL

eg: cats → cat

Step 1b Rules:-

Removes eed, ed, ing

ex:- agreed → agree

Making → Make

Step 1c Rule:- happy → happi

cry → cri

letter y becomes i

Step 2 rules: Replace longer suffixes

ex:- ational
↓
ate

relational → relate

Step 3 rules:- icate → ic

duplicate → duplic

Step 4 rules: Removes suffixes such as

ance

ence

ment

able

ible

ion

en

ic

adjustable → adjust

Step 5: final connections.

control
↓
control

duplicate
↓
duplicate
↓
duplicate

Sometimes the ending
e is removed.

Dictionary look-up stemmer is a stemming technique that uses a predefined dictionary (lexicon) to find the correct root (stem) of a word instead of depending only on suffix-removal rules.

unlike, simple stemming algorithms (such as Porter stemmer) — this method checks whether the resulting word actually exists in the dictionary before accepting it as a stem.

Dictionary look-up stemmer is a stemming technique in which every word is searched in a dictionary and the valid root word stored in the dictionary is returned as the stem.

Need for dictionary look-up stemmer:
Simple stemming algorithms remove suffixes based on fixed rules.

ex:- Playing — play
Working — work

But sometimes rule based algorithms produce incorrect stems.

ex:- university — univers
memorial — memor

To avoid such mistakes, dictionary look-up stemmers verify the stem using a dictionary.

Working principle: The process works as follows.

input word



Apply simple stemming Rules



search dictionary



if root word exists



return root word.



else keep original word.

ex 1: input
running
↓
Rule removes "ing"
↓
run
↓
Dictionary contains "Run"
↓
output = Run

ex 2: input
university
↓
Rule remove suffix
↓
universe
↓
Dictionary doesn't contain "univers"
output = university.

KStem (Krovetz Stemmer)

The INQUERY Retrieval system uses a dictionary based Stemmer called KStem (Krovetz Stemmer)

KStem performs Morphological analysis & Reduces different word forms into a common root while avoiding incorrect stemm-
-ing.

Krovetz Stemmer is a dictionary-based stemming algorithm developed by Robert Krovetz (1993)

Instead of simply removing suffixes, it verifies whether the resulting word is a valid English word.

KStem's main working step is dictionary lookup.

the process: -
input word
↓
Remove simple suffix (if possible)
↓
lookup the resulting word in the dictionary
if found → Accept it as a stem

→ keep the original word (or) apply another valid rule.

Characteristics of Kstem

- uses dictionary
- returns Real Words
- avoid over-stemming
- handles irregular words
- perform morphological analysis

Six major data files used by Kstem.
- the Kstem system controls stemming using six important files.

1) Dictionary (lexicon)

Stores all valid English words

ex:- Run
Play
Study
Computer
Memory

2) Supplemental Dictionary :

Contains newly added words.

ex:- internet
Smartphone
Blockchain
chatGPT

3) Exception list: Stores words that should not follow normal stemming rules.

ex:- suites → suite
suited → suit

4) Direct conflation list:

Maps one word directly to another

ex:- children → child
people → person

5) Country - Nationality list:

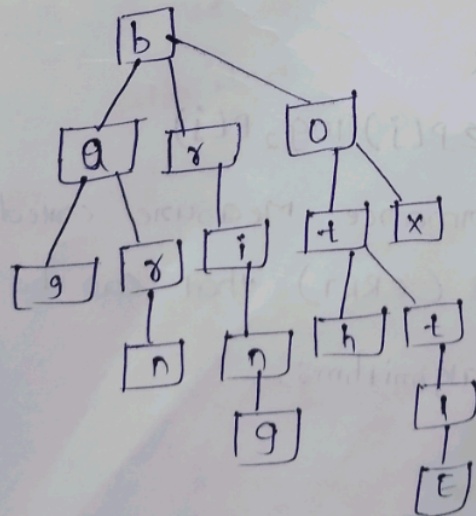
Maps nationality names to country names

ex:- British → Britain
Indian → India
Japanese → Japan

Rama

Bottle

symbol tree



Successor Stemmen divides words

1) cutoff Method:
A fixed cutoff value is selected to determine stem length

ex: computing
↓
cut after "comput"
↓
stem = comput

2) peak and plateau Method
Break the word where successor variety reaches its highest value

3) complete word Method:
Break only at complete meaningful words

ex: Blackboard
↓
Black + board

4) entropy Method: uses Shannon's entropy to find the most informative position for splitting the word.
↓
the position in a word where splitting it gives the maximum information about the stem & suffix.

Higher entropy indicates a better stem boundary.

Formula:

Probability of successor = $P[j]$

Words with the same prefix = D_k

Words having successor $j = D_k j$

$$P[j] = D_k j / D_k$$

$$\text{Entropy}(H) = -\sum P(j) \log_2 P(j)$$

a stemming performance measure called error rate
Relative to truncate (ERRT) that can be used to
compare stemming algorithms.

information extraction:

information extraction is the process of automatically extracting useful and specific information from text doc's & storing it in structured form.

Why is information extraction needed:

- * to extract information facts from large collection of documents
- * to reduce manual effort
- * to improve search accuracy in IRS
- * to organize unstructured data into structured databases

How information extraction works.

1) input: ~~un~~ unstructured text document.

2) Text Analysis: the system analyzes the document.

3) Entity identification: finds names of people, places, organizations, dates etc..

4) Relationship extraction: identifies relationships b/w entities.

5) Structured output: stores the extracted information in a database or table

ex:- Satya Nadella because the CEO of Microsoft in 2014

extracted info

entity value

person

Satya Nadella

organization

Microsoft

Position

CEO

Year

2014.

inverted file structure:-

Introduction:- the inverted file structure is the most commonly used data structure in both database management systems and information retrieval system (IRS)

it is designed to retrieve documents quickly by storing each word (term) along with the list of documents in which that word appears.

unlike sequential searching, where every document is scanned one by one the inverted file structure directly identifies the required documents, making the search process faster and more efficient.

Definition:- An inverted file structure is a technique used to store words instead of documents. each word is connected to the list of document numbers where it appears.

Why is it called an "inverted" file?

normally, documents are stored like this normal file.

DOC1 → computer, Bit, Byte

DOC2 → memory, Byte

DOC3 → computer, Memory, Bit

DOC4 → Byte, computer.

Here

document → words

this is called the forward file structure.

* in an inverted file structure, the storage is inverted/reversed.

computer → DOC1, DOC3, DOC4

Bit → DOC1, DOC3

Byte → DOC1, DOC2, DOC4

Memory → DOC2, DOC3

Now

word → documents.

since the storage direction is reversed (inverted)

Need for inverted file structure:

When thousands or millions of documents are stored, searching each document one by one becomes very slow

ex:- suppose Google store 100 crore webpages

if a user searches "Artificial intelligence",

Google cannot open and read every webpage

⇒ instead, it uses an inverted file structure

the dictionary directly finds the word "Artificial", then retrieves the list of webpage IDs containing that word.

this makes searching, extremely fast.